

suggest that controlled retrieval depends on flexible gating of perceptual and mnemonic features according to current cognitive demands. Experiments 3 and 4 demonstrated the functional significance of this architecture. In resting-state data, stronger intrinsic coupling between the interaction architecture and visual cortex predicted better reading comprehension and reduced mind-wandering propensity. In a conflict paradigm manipulating concurrent reading and autobiographical retrieval, the same architecture showed enhanced visual coupling when perceptual representations needed to be stabilised against internally generated interference. Together, these findings suggest that declarative memory is organised not simply according to memory content, but through a frontoparietal–DMN interaction architecture that dynamically coordinates perceptual and mnemonic representations across internally and externally oriented cognition.

E9 Sandbox Series - Bilingual Language Context Modulates Neural Indices of Executive Control in Cantonese–English Bilinguals

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The relationship between bilingual language control and domain-general executive control has often been examined through cross-task conflict adaptation. Namely, the control demands induced by a preceding language task may modulate performance on a subsequent non-linguistic task. Previous work with highly balanced Welsh–English early bilinguals showed that a mixed-language context reduced the Flanker congruency effect, suggesting enhanced executive control after language switching. However, a similar study testing Mandarin–English late bilinguals, whose L2-English proficiency was much lower (scored 3.8 out of 7) than L1-Mandarin, did not find the same effect. The mixed findings may partly reflect differences in bilingual profile (i.e., L2 proficiency). Therefore, we tested a more balanced, higher-L2 proficiency group, that is, Cantonese–English bilinguals in Hong Kong. They are more balanced between their two languages while maintaining a language-distance profile broadly comparable to that of Mandarin–English bilinguals. The participants, with self-rated English proficiency averaged at 5.14 on a 1–7 scale and a mean age of English acquisition of 3.1 years, completed three counterbalanced language-context blocks: Cantonese-single, English-single, and mixed. Each block began with an auditory picture–word matching task (64 trials) designed to establish the language context, followed immediately by a 64-trial Flanker task. In the language-context phase, participants judged whether a word matched a visually presented picture. Single-language conditions contained words from only one language, whereas the mixed-language conditions contained 32 Cantonese and 32 English words (pseudo-randomly intermixed). In the Flanker task, participants identified the direction of a central arrow flanked by congruent or incongruent arrows. Reaction times in the Flanker were analysed. EEG was recorded during the Flanker task, with analyses focusing on P3 (350–700 ms) amplitude at centro-parietal electrodes. While the data collection is going on, the behavioural results from 45 participants showed robust Flanker congruency effects in all language conditions ($\beta = 0.01$, $SE = 0.07$, $t = 14.13$, $p < .001$), with slower responses to incongruent than to congruent trials. This congruency effect was

numerically the smallest after the mixed condition. The EEG results provided robust evidence for language-context modulation. The P3 amplitude showed significant congruency difference after both Cantonese-single ($\beta = 4.69$, $SE = 0.66$, $z = 7.08$, $p < .001$) and English-single conditions ($\beta = 1.60$, $SE = 0.68$, $z = 2.34$, $p = .019$), whereas this difference was absent after the mixed condition ($\beta = -0.41$, $SE = 0.66$, $z = -0.62$, $p = .532$). The results showed that the mixed condition reduced neural sensitivity to conflict in the Flanker task, compared with single-language conditions. For Cantonese–English bilinguals, the mixed condition pre-engaged executive control mechanisms, and this control state carried over to subsequent Flanker tasks. The current sample of bilinguals had relatively homogeneous L2 proficiency, such that variability in proficiency did not produce detectable differences in cross-task adaptation in the statistical analysis. While more participants with a broader proficiency range will be tested, the findings reconcile previous mixed evidence by suggesting that the language context effect on executive control may be modulated by bilingual language profiles.

E10 - Immersion affects attentional processing in bilinguals

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Introduction: Bilingualism modulates neural mechanisms of auditory selective attention: bilinguals often organize mental resources more flexibly under challenging speech-processing scenarios while maintaining optimal comprehension (Olguin et al., 2019; Phelps et al., 2022). This reflects neuroplastic adaptation to an environment where target-language selection and non-target inhibition create competition and increased cognitive demands (Kroll & Bialystok, 2013). However, bilingualism is not a binary concept, but a complex spectrum of various facets (Yow & Li, 2015). Factors such as immersion in an L2-dominant environment may further fine-tune bilinguals' cognitive processing (e.g., Nicolay & Poncelet, 2013), yet such influence remains under-explored. This study aimed to investigate how immersion in an L2 context impacts attentional processing of speech. Methods: Participants were native Mandarin Chinese speakers who learned English by the age of six: 31 in Beijing (without English immersion) and 27 in Cambridge, UK (with English immersion), matched for age, L2-acquisition age and proficiency. In a dichotic listening task simulating real-life scenarios, participants attended to a naturalistic, continuous Chinese narrative in one ear while simultaneously ignoring a competing stream in the other. The type of interference was manipulated to create four conditions featuring distractors of varying intelligibility: (1). Single Talker (ST; no interference), (2). Native-Musical Rain (MuR, non-linguistic interference), (3). Native-Unknown (unintelligible linguistic interference – narratives in Serbian), and (4). Native-Native (intelligible linguistic interference – narratives in Chinese). EEG was recorded, pre-processed and band-filtered to 1–12Hz. Neural tracking of stimuli was analyzed by mapping between speech envelopes and neural responses using backward temporal-response-function (mTRF). Linear mixed-effects models assessed effects of immersion group, conditions, attention and L2-immersion years on neural tracking of stimuli. Results: Behavioral comprehension was uniformly high and neural decoding patterns were similar in two groups. Analysis